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STRAIGHT LINE
NIMCET PYQ

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NIMCET-Free Test



By Amit Katiyar (MCA-JNU)



- The distance between the pair of straight line $x^2 + 2\sqrt{3}xy + 2y^2 + 4x + 1 = 0$
(a) $2\sqrt{3}$ (b) 2 (c) $\sqrt{3}$ (d) None
NIMCET 2007
- The product of perpendiculars from $(2, -1)$ to the pair of lines $2x^2 + 6xy + y^2 = 0$ is:
(a) $\frac{3}{2}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{3}{37}$ (d) $\frac{3}{\sqrt{37}}$
NIMCET 2007
- If $y = mx$ bisects the angle between the lines $x^2(\tan^2 \theta + \cos^2 \theta) + 2xy \tan \theta - y^2 \sin^2 \theta = 0$ when $\theta = \frac{\pi}{3}$, then the value of $\sqrt{3}m^2 + 4m$ is
(a) 1 (b) $1/\sqrt{3}$ (c) $\sqrt{3}$ (d) $7/\sqrt{3}$
NIMCET 2008
- A line L has intercepts 'a' and 'b' on the coordinates axes. When the axes are rotated through a given angle, keeping the origin fixed, the same line has intercepts 'p' and 'q' which of the following statements is true?
(a) $a^2 + b^2 = p^2 + q^2$ (b) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
(c) $a^2 + p^2 = b^2 + q^2$ (d) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$
NIMCET 2008
- If a, b, c are the roots of the equation $x^3 - 3px^2 + 3qx - 1 = 0$, then the centroid of the triangle with vertices $(a, \frac{1}{a})$, $(b, \frac{1}{b})$ and $(c, \frac{1}{c})$ is at the point
(a) (p, q) (b) $(\frac{p}{3}, \frac{q}{3})$
(c) $(p + q, p - q)$ (d) $(3p, 3q)$
NIMCET 2008
- If the distance of any point (x, y) from the origin is defined as $d(x, y) = \max(|x|, |y|)$, then the locus of the point (x, y), where $d(x, y) = 1$ is
(a) a square of area 1 sq. unit
(b) a circle of radius 1 sq. unit
(c) a triangle
(d) a square of area 4 sq. units
NIMCET 2009
- Let ABC be an isosceles triangle with $AB = BC$. If base BC is parallel to x-axis and m_1, m_2 are slopes of medians drawn through the angular points B and C then,
(a) $m_1 m_2 = -\frac{1}{2}$
(b) $m_1 + m_2 = 0$
(c) $m_1 - m_2 = 2$
(d) $(m_1 + m_2)^2 + 2m_1 m_2 = 0$
NIMCET 2009
- The straight lines $\frac{x}{a} - \frac{y}{b} = k$ and $\frac{x}{a} + \frac{y}{b} = \frac{1}{k}, k \neq 0$ meet on
(a) a parabola (b) an ellipse
(c) a hyperbola (d) a circle
NIMCET 2009
- The distance between the parallel lines $y = 2x + 4$ and $6x = 3y + 5$;
(a) $\frac{17}{\sqrt{3}}$ (b) 1 (c) $\frac{3}{\sqrt{5}}$ (d) $\frac{17\sqrt{5}}{15}$
NIMCET 2010

- The angle of intersection of the cardioids $r = a(1 + \cos \theta), r = a(1 - \cos \theta)$ is
(a) $\frac{\pi}{2}$ (b) 0 (c) $\frac{\pi}{4}$ (d) π
NIMCET 2010
- If the two pairs of lines $x^2 - 2mxy - y^2 = 0$ and $x^2 - 2nxy - y^2 = 0$ are such that one of them represent the bisector of the angles between the other, then
(a) $mn + 1 = 0$ (b) $mn - 1 = 0$
(c) $\frac{1}{m} + \frac{1}{n} = 0$ (d) $\frac{1}{m} - \frac{1}{n} = 0$
NIMCET 2010
- The point (4, 1) undergoes the following three transformations successively
(i) reflection about the line $y = x$
(ii) transformation through a distance 2 unit along the positive directions of x-axis
(iii) rotation through an angle of $\frac{\pi}{4}$ about the origin in the anti-clock wise direction. The final positions of the point is given by the coordinates.
(a) $(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}})$ (b) $(\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}})$
(c) $(-2, 7\sqrt{2})$ (d) $(\sqrt{2}, 7\sqrt{2})$
NIMCET 2010
- The number of values of k for which the system of equations $(k + 1)x + 8y = 4k$ and $kx + (k + 3)y = 3k - 1$ has infinitely many solutions, is
(a) 0 (b) 1 (c) 2 (d) infinite
NIMCET 2012
- If (4, -3) and (-9, 7) are the two vertices of a triangle and (1, 4) is its centroid, then the area of triangle is
(a) $\frac{138}{2}$ (b) $\frac{319}{2}$ (c) $\frac{183}{2}$ (d) $\frac{381}{2}$
NIMCET 2012
- The equation of the line parallel to the line $2x - 3y = 7$ and passing through the mid-point of the line segment joining the points (1, 3) and (1, -7) is
(a) $2x - 3y - 4 = 0$ (b) $2x - 3y + 4 = 0$
(c) $2x - 3y - 8 = 0$ (d) $2x - 3y + 8 = 0$
NIMCET 2013
- The equation of the base of an equilateral triangle is $x + y = 2$ and the vertex is (2, -1). The length of the side of the triangle is
(a) $\sqrt{\frac{3}{2}}$ (b) $\sqrt{2}$ (c) $\sqrt{\frac{2}{3}}$ (d) $\sqrt{\frac{20}{3}}$
NIMCET 2013
- The foot of the perpendicular from the point (2, 4) upon $x + y = 1$ is
(a) $(\frac{1}{2}, \frac{3}{2})$ (b) $(-\frac{1}{2}, \frac{3}{2})$ (c) $(\frac{4}{3}, \frac{1}{2})$ (d) $(\frac{4}{3}, -\frac{1}{2})$
NIMCET 2015
- If (2, 1), (-1, -2), (3, 3) are the mid-points of the sides BC, CA and AB of a triangle ABC, then equation of the line BC is
(a) $5x + 4y + 6 = 0$ (b) $5x - 4y - 6 = 0$
(c) $5x + 4y - 6 = 0$ (d) $5x - 4y + 6 = 0$
NIMCET 2015



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19. If $(-4, 5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the other diagonal is

NIMCET 2015

- (a) $x + 7y = 21$ (b) $x + 7y = 31$
(c) $x - 7y = 31$ (d) $x + 7y = 35$

20. If $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$ represents pair of straight lines, the value of λ is

NIMCET 2016

- (a) 2 (b) 4 (c) 6 (d) 8

21. If the lines $x + (a - 1)y + 1 = 0$ and $2x + a^2y - 1 = 0$ are perpendicular, the condition satisfied by a is

NIMCET 2017

- (a) $|a| = 2$ (b) $0 < a < 1$
(c) $-1 < a < 0$ (d) $a = -1$

22. If $x^2 + 3xy + 2y^2 - x - 4y - 6 = 0$ represents a pair of straight lines, their point of intersection is

NIMCET 2017

- (a) $(0, 0)$ (b) $(8, 5)$ (c) $(8, -5)$ (d) $(-2, 5)$

23. The locus of the orthocentre of the triangle formed by the lines $(1 + p)x - py + p(1 + p) = 0$, $(1 + q)x - qy + q(1 + q) = 0$ and $y = 0$ where $p \neq q$ is

NIMCET 2018

- (a) a hyperbola (b) a parabola
(c) an ellipse (d) a straight line

24. Equation of the line perpendicular to $x - 2y = 1$ and passing through $(1, 1)$ is

NIMCET 2018

- (a) $x + 2y = 3$ (b) $x + y = 2$
(c) $y = 2x + 3$ (d) $y = -2x + 3$

25. A line through $(4, 2)$ meets the coordinate axes at P and Q. then, the locus of the circumcentre of ΔOPQ is

NIMCET 2018

- (a) $\frac{1}{x} + \frac{1}{y} = 2$ (b) $\frac{2}{x} + \frac{1}{y} = 1$
(c) $\frac{1}{x} + \frac{2}{y} = 1$ (d) $\frac{1}{x} + \frac{1}{y} = \frac{1}{2}$

26. Slope of two lines $6x^2 - 2xy - 2y^2 = 0$ differ by

NIMCET 2018

- (a) $\frac{5}{2}$ (b) $\frac{7}{2}$ (c) $\frac{7}{5}$ (d) -1

27. If θ is the acute angle between the pair of lines $x^2 - 2xy + 12y^2 = 0$, then $\frac{2 \cos \theta + 3 \sin \theta}{4 \sin \theta + 5 \cos \theta} = ?$

NIMCET 2021

- (a) $\frac{29}{69}$ (b) $\frac{61}{39}$ (c) $\frac{39}{61}$ (d) $\frac{69}{29}$

28. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram PQRS, then

NIMCET 2021

- (a) $a = 2, b = 3$ (b) $a = 3, b = 4$
(c) $a = 2, b = 4$ (d) $a = 3, b = 5$

29. The lines $px + qy = 1$ and $qx + py = 1$ are respectively the sides AB, AC of the triangle ABC and the base BC is bisected at (p, q) . equation of median of the triangle through the vertex A is

NIMCET 2021

- (a) $(2pq - 1)(qx + py - 1) + (p^2 + q^2 - 1)(px + qy - 1) = 0$
(b) $(2pq - 1)(qx + py - 1) - (p^2 + q^2 - 1)(px + qy - 1) = 0$
(c) $(2pq - 1)(px + qy - 1) + (p^2 + q^2 - 1)(qx + py - 1) = 0$
(d) $(2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)(qx + py - 1) = 0$

30. Area of parallelogram formed by the lines $y = 4x$, $y = 4x + 1$, $x + y = 0$ and $x + y = 1$ is

NIMCET 2022

- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$ (c) 5 (d) 10

31. A straight line through the point $(4, 5)$ is such that its intercepts between the axis is bisected at A, then its equation is

NIMCET 2022

- (a) $3x + 4y = 20$ (b) $3x - 4y + 7 = 0$
(c) $5x - 4y = 40$ (d) $5x + 4y = 40$

32. Let a be the distance between lines $-2x + y = 2$ and $2x - y = 2$, and b be the distance between the lines $4x - 3y = 5$ and $6y - 8x = 1$, then

NIMCET 2022

- (a) $40b = 11\sqrt{5}a$ (b) $40\sqrt{2}a = 11b$
(c) $11\sqrt{2}b$ (d) $11\sqrt{2}a = 40b$

33. Let a, b, c, d be non-zero numbers. If the point of intersection of the line $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistance from the two are then

NIMCET 2023

- (a) $a + b + c + d = 0$ (b) $ad - bc = 0$
(c) $3bc - 2ad = 0$ (d) $3bc + 2ad = 0$

34. The range of values of θ in the interval $(0, \pi)$ such that the points $(3, 2)$ and $(\cos \theta, \sin \theta)$ lie on the same sides of the line $x + y - 1 = 0$, is

NIMCET 2023

- (a) $(0, \frac{3\pi}{4})$ (b) $(0, \frac{\pi}{2})$
(c) $(0, \frac{\pi}{3})$ (d) $(0, \frac{\pi}{4})$

35. If the perpendicular bisector of the line segment joining $p(1, 4)$ and $q(k, 3)$ has y-intercept -4 , then the possible value of k are (Straight Line)

NIMCET 2024

- (a) -4 and 4 (b) -2 and 2
(c) -3 and 3 (d) -1 and 1

36. The points $(1, 1/2)$ and $(3, -1/2)$ are (Straight line)

- (a) In between the lines $2x + 3y = 6$ and $2x + 3y = -6$
(b) on the same side of the line $2x + 3y = 6$
(c) on the same side of the line $2x + 3y = -6$
(d) on the opposite side of the line $2x + 3y = -6$

NIMCET 2024





37. For what values of λ does the equation $6x^2 - xy + \lambda y^2 = 0$ represents two perpendicular lines and two lines inclined at an angle of $\pi/4$. (Pair of Lines)
- (a) -6 and -2 (b) 6 and 1
(c) -1 and -35 (d) -6 and 1

NIMCET 2024

38. The obtuse angle between lines $2y = x + 1$ and $y = 3x + 2$ is

NIMCET 2025

- (a) $\frac{3\pi}{4}$ (b) $\frac{5\pi}{6}$
(c) $\frac{4\pi}{3}$ (d) $\frac{2\pi}{3}$

39. Let the line $\frac{x}{4} + \frac{y}{2} = 1$ meets the x-axis and y-axis at A and B, respectively. M is the midpoint of side AB, and M' is the image of the point M across the line $x + y = 1$. Let the point P lie on the line $x + y = 1$ such that the ΔABP is an isosceles triangle with $AP = BP$. Then the distance between M' and P is

NIMCET 2025

- (a) $\frac{\sqrt{7}}{3}$ (b) $\frac{2\sqrt{5}}{3}$
(c) $\frac{2\sqrt{7}}{3}$ (d) $\frac{\sqrt{5}}{3}$

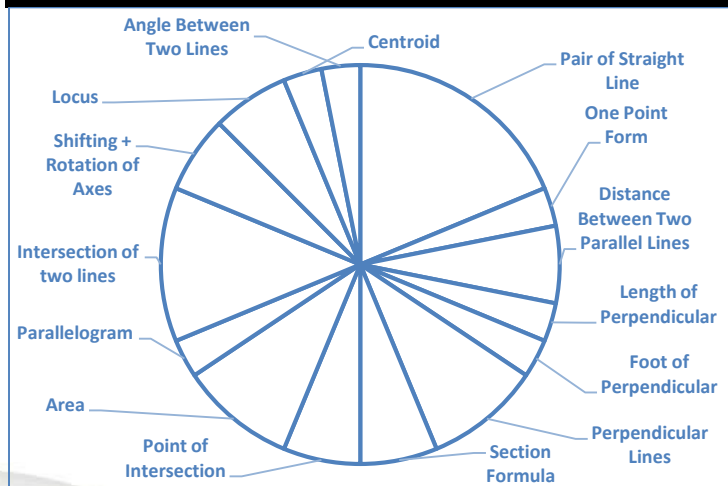
ANSWER KEY

1.	b	2.	d	3.	c	4.	b	5.	a
6.	d	7.	b	8.	c	9.	d	10.	a
11.	a	12.	a	13.	b	14.	c	15.	c
16.	c	17.	b	18.	b	19.	b	20.	d
21.	d	22.	c	23.	d	24.	d	25.	b
26.	b	27.	a	28.	a	29.	d	30.	a
31.	d	32.	a	33.	c	34.	b	35.	a
36.	a	37.	c	38.	a	39.	b		

Analysis 2007 to 2025 Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	02	NIMCET 2016	01
NIMCET 2008	03	NIMCET 2017	02
NIMCET 2009	03	NIMCET 2018	04
NIMCET 2010	04	NIMCET 2019	00
NIMCET 2011	00	NIMCET 2020	00
NIMCET 2012	02	NIMCET 2021	03
NIMCET 2013	02	NIMCET 2022	03
NIMCET 2014	00	NIMCET 2023	02
NIMCET 2015	03	NIMCET 2024	03
		NIMCET 2025	02

SUB TOPICWISE ANALYSIS



SUB TOPICWISE ANALYSIS

Sub Topic	Qs. Asked	Sub Topic	Qs. Asked
Pair of Straight Line	06	Area	03
One Point Form	01	Parallelogram	01
Distance Between Two Parallel Lines	02	Intersection of two lines	04
Length of Perpendicular	01	Shifting + Rotation of Axes	02
Foot of Perpendicular	01	Locus	02
Perpendicular Lines	03	Centroid	01
Section Formula	02	Angle Between Two Lines	01
Point of Intersection	02		

SOLUTION

1. (b) $x^2 + 2\sqrt{3}xy + 2y^2 + 4x + 1 = 0$
Distance between pair of str. Line $-2\sqrt{\frac{g^2 - ac}{a(a+b)}}$

$$\Rightarrow 2\sqrt{\frac{(2)^2 - 1 \times 1}{1(1+2)}} \\ 2\sqrt{\frac{3}{3}} = 2$$

2. (d)
Product of perpendicular from $(2, -1)$ to pair of line $2x^2 + 6xy + y^2 = 0$

Product of perpendicular from (x_1, y_1) to pair of line

$$ax^2 + by^2 + 2bxy + ax + 2fy + c \text{ is} \\ = \frac{a(x_1)^2 + b(y_1)^2 + 2hx_1y_1 + 2gx_1 + 2y_1 + c}{\sqrt{(a-b)^2 + 4h^2}}$$

$$\Rightarrow \frac{2(2)^2 + c(2)(-1) + (-1)^2}{\sqrt{(2-1)^2 + 4 \times (3)^2}} \\ = \frac{|8 - 12 + 1|}{\sqrt{37}} = \frac{-3}{\sqrt{37}} = \frac{3}{\sqrt{37}}$$

3. (c)

Given lines,

$$x^2(\tan^2 \theta + \cos^2 \theta) + 2xy \tan \theta - y^2 \sin^2 \theta = 0$$

$$\text{Its angle bisector } \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

$$\frac{x^2 - y^2}{\tan^2 \theta + \cos^2 \theta - (-\sin^2 \theta)} = \frac{xy}{\tan \theta}$$



$$\Rightarrow \frac{x^2 - y^2}{\sec^2 \theta} = \frac{xy}{\tan \theta}$$

$$\text{At } \theta = \frac{\pi}{3}, \frac{x^2 - y^2}{4} = \frac{xy}{\sqrt{3}}$$

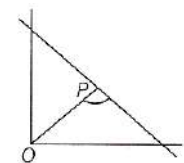
Also, $y = mx$ satisfies Eq. (i)

$$\frac{x^2 - m^2 x^2}{4} = \frac{x^2 m}{\sqrt{3}}$$

$$\Rightarrow \sqrt{3}(1 - m^2) = 4m$$

$$\Rightarrow \sqrt{3}m^2 + 4m = \sqrt{3}$$

4. (b) Before rotation of axes



Equation of lines, $\frac{x}{a} + \frac{y}{b} = 1$ (i)

After rotation,

Equation of lines, $\frac{x}{p} + \frac{y}{q} = 1$ (ii)

But in both case perpendicular distance from origin will remain same

$$\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2}}} \Rightarrow \frac{1}{p^2} + \frac{1}{q^2} = \frac{1}{a^2} + \frac{1}{b^2}$$

5. (a) $x^3 - 3px^2 + 3qx - 1 = 0$

Its roots are a, b and c

Here we use Vieta's theorem

$$a + b + c = 3p, ab + bc + ac = 3q, abc = 1$$

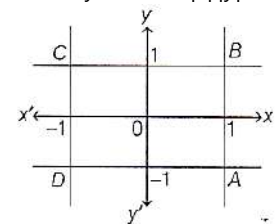
$$\left(a, \frac{1}{a}\right), \left(b, \frac{1}{b}\right), \left(c, \frac{1}{c}\right)$$

$$\text{Centroid } (x, y) = \left(\frac{a+b+c}{3}, \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3}\right)$$

$$= \left(\frac{3p}{3}, \frac{1}{3} \left(\frac{ab+bc+ac}{abc}\right)\right)$$

$$= \left(p, \frac{1}{3} \times \frac{3q}{1}\right) = (p, q)$$

6. (d) $d(x, y) = \max(|x|, |y|) = 1$



If $x > y, |x| = 1 \Rightarrow x = \pm 1$, If $y > x, |y| = 1 \Rightarrow y = \pm 1$

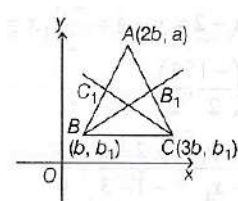
$\therefore$ We get a square ABCD on plotting the above lines.

Side length of square = 2 units.

$\therefore$ Area of square = 4 sq. units

$\therefore$ Locus of point (x, y) is a square of area 4 sq. units.

7. (b) As we know ABC is an isosceles triangle and BC is parallel to x-axis.



Let $B = (b, b_1), C = (3b, b_1)$ Hence, $A = (2b, a)$ B_1 is the mid point of AC, hence BB_1 is median of AC, C_1 is the mid point of AB, hence CC_1 is median of AB, $B_1 = \left(\frac{5b}{2}, \frac{a+b_1}{2}\right), C_1 = \left(\frac{3b}{2}, \frac{a+b_1}{2}\right)$

$$\text{Slope of } BB_1(m_1) = \frac{\frac{a+b_1}{2} - b_1}{\frac{5b}{2} - b} = \frac{a-b_1}{3b}$$

$$\text{Slope of } CC_1(m_1) = \left(\frac{\frac{a+b_1}{2} - b_1}{\frac{3b}{2} - 3b}\right) = \frac{a-b_1}{-3b}$$

Hence, $m_1 + m_2 = 0$

8. (c) $\frac{x}{a} - \frac{y}{b} = k, \frac{x}{a} + \frac{y}{b} = \frac{1}{k}, k \neq 0$

Since, both lines are intersecting

$$\therefore \text{Pair of equation of lines, } \left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = k \times \frac{1}{k}$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Which is a hyperbola.

9. (d) Distance between parallel lines.

$$2x - y + 4 = 0 \Rightarrow 6x - 3y - 5 = 0 \Rightarrow 2x - y - \frac{5}{3} = 0$$

To calculate distance between parallel lines coefficient of x, y should be same

$$D = \left| \frac{c_1 - c_2}{\sqrt{a^2 + b^2}} \right| = \left| \frac{4 - \left(-\frac{5}{3}\right)}{\sqrt{4 + 1}} \right| = \frac{17}{3\sqrt{5}} = \frac{17\sqrt{5}}{15} \text{ units}$$

10. (a) $r = a(1 + \cos \theta), r = a(1 - \cos \theta)$

For angle of intersection

$$a + a \cos \theta = a - a \cos \theta \Rightarrow 2a \cos \theta = a - a$$

$$\Rightarrow 2a \cos \theta = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

11. (a) $ax^2 + 2bxy + by^2 = 0$

$$\text{Equation of angle bisector, } \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$$

Given, $x^2 - 2mxy - y^2 = 0$ is angle bisector of

$$x^2 - 2nxy - y^2 = 0 \quad \text{..... (i)}$$

$$\frac{x^2 - y^2}{1 - (-1)} = \frac{xy}{-m}$$

$$-mx^2 + my^2 = 2xy$$

$$-mx^2 + my^2 - 2xy = 0 \quad \text{..... (ii)}$$

$\therefore$ Eqs. (i) and (ii) are same

$$\therefore \frac{-m}{1} = \frac{m}{-1} = \frac{-2}{-2n}$$

$$\Rightarrow 2mn + 2 = 0 \Rightarrow mn + 1 = 0$$

12. (a) P (4, 1)

(i) reflection about the line $y = x$

$P' (1, 4)$

(ii) transformation through a distance 2 unit along the positive directions of x -axis

$P' (3, 4)$

(iii) rotation through an angle of $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3\cos \frac{\pi}{4} - 4\sin \frac{\pi}{4} \\ 3\sin \frac{\pi}{4} + 4\cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ \frac{7}{\sqrt{2}} \end{bmatrix}$$

$$\therefore x' = \frac{-1}{\sqrt{2}}, y' = \frac{7}{\sqrt{2}}$$

$$\text{Required point} = \left(\frac{-1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$$



- 13. (b)** Condition that 2 lines are coincident

$$a_1x + b_1y + c_1 = 0 \Rightarrow a_2x + b_2y + c_2 = 0$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow (k+1)x + 8y = 4k$$

$$\Rightarrow kx + (k+3)y = 3k-1 \Rightarrow \frac{k+1}{k} = \frac{8}{k+3} = \frac{4k}{3k-1}$$

$$k^2 + 4k + 3 = 8k; 24k - 8 = 4k^2 + 12k$$

$$k^2 - 4k + 3 = 0; 4k^2 - 12k + 8 = 0$$

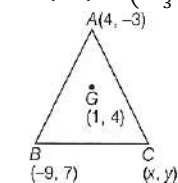
$$(k-3)(k-1) = 0; k^2 - 3k + 2 = 0$$

$$k = 3, k = 1 \text{ and } k = 1, 2.$$

$\therefore$ For $k = 1$ the lines will have infinite solution.

$\therefore$ Number of values of $k = 1$.

- 14. (c)** $(1, 4) = \left(\frac{4-9+x}{3}, \frac{-3+7+y}{3}\right)$

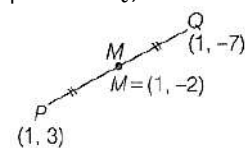


$$x = 8, y = 8$$

$$\text{area}(\triangle ABC) = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\text{area}(\triangle ABC) = \frac{1}{2} \begin{vmatrix} 4 & -3 & 1 \\ -9 & 7 & 1 \\ 8 & 8 & 1 \end{vmatrix} = \frac{183}{2}$$

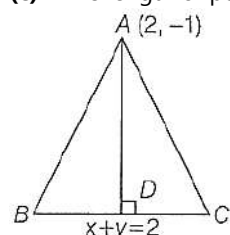
- 15. (c)** Let equation of line, $2x - 3y = \lambda$ passing through M (mid point of PQ)



$$2(1) - 3(-2) = \lambda \Rightarrow \lambda = 8$$

$$\Rightarrow 2x - 3y = 8 \Rightarrow 2x - 3y - 8 = 0$$

- 16. (c)** AD is length of perpendicular from vertex A to base BC .



$$\text{In } \triangle ABD, \tan 60^\circ = \frac{AD}{a/2} \Rightarrow AD = \frac{\sqrt{3}a}{2}$$

Where a is side of equilateral triangle

$$AD = \left| \frac{2-1-2}{\sqrt{1+1}} \right| = \frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{2}a \Rightarrow a = \sqrt{\frac{2}{3}}$$

- 17. (b)** Foot of perpendicular

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = -\frac{(ax_1+by_1+c)}{a^2+b^2}$$

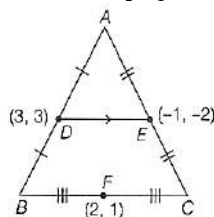
$$\text{Here, } (x_1, y_1) = (2, 4)$$

$$\frac{x-2}{1} = \frac{y-4}{1} = \frac{-(2+4-1)}{1+1}$$

$$x-2 = y-4 = \frac{-5}{2}, x = \frac{-1}{2}, y = \frac{3}{2}$$

$$\therefore B\left(\frac{-1}{2}, \frac{3}{2}\right)$$

- 18. (b)** $m_{DE} = \frac{y_2-y_1}{x_2-x_1} = \frac{-2-3}{-1-3} = \frac{5}{4}$



$DE \parallel BC$: equation of $BC = y = mx + c$

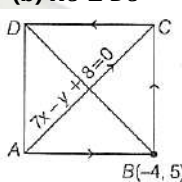
$$y = \frac{5}{4}x + c$$

$$\text{Also, } F \text{ lies on } BC: 1 = \frac{5}{4} \times 2 + c \Rightarrow c = 1 - \frac{5}{2}, c = -\frac{3}{2}$$

$$\therefore BC: y = \frac{5}{4}x - \frac{3}{2}$$

$$4y = 5x - 6 \Rightarrow 5x - 4y - 6 = 0$$

- 19. (b)** $AC \perp BC$



$$\therefore m_{AC} = 7$$

$$\therefore m_{BD} = -\frac{1}{7}$$

$$BD: y = -\frac{1}{7}x + c$$

$$(-4, 5) \text{ lies on } BD: 5 = \frac{4}{7} + c, c = \frac{31}{7}$$

$$\therefore BD: y = -\frac{1}{7}x + \frac{31}{7} \Rightarrow x + 7y = 31$$

- 20. (d)** $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$

For pair of straight line $\Delta = 0$

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$a = 2, b = 3, h = \frac{7}{2}, g = 4, f = 7, c = \lambda$$

$$6\lambda + 2 \times \frac{7}{2} \times 4 \times 7 - (2 \times 49) - (3 \times 16) - \left(\lambda \times \frac{49}{4}\right) = 0$$

$$\Rightarrow 6\lambda + 196 - 146 - \frac{49\lambda}{4} = 0 \Rightarrow \frac{24\lambda - 49\lambda}{4} + 50 = 0$$

$$\Rightarrow -25\lambda = -200 \quad \lambda = 8$$

- 21. (d)** $L_1: x + (a-1)y + 1 = 0, L_2: 2x + a^2y - 1 = 0$

$$m_{L_1} = \frac{1}{1-a}, m_{L_2} = \frac{-2}{a^2}$$

$$m_{L_1} \times m_{L_2} = -1 \Rightarrow \frac{1}{1-a} \times \frac{2}{a^2} = 1$$

$$\Rightarrow 2 = a^2(1-a) \Rightarrow a^3 - a^2 + 2 = 0$$

$$\Rightarrow (a+1)(a^2 - 2a + 2) = 0 \Rightarrow (a+1) = 0; a^2 - 2a + 2 = 0$$

$$a = -1$$

- 22. (c)** $f: x^2 + 3xy + 2y^2 - x - 4y - 6 = 0$

$$\frac{\partial f}{\partial x} = 2x + 3y - 1 = 0$$

$$\frac{\partial f}{\partial y} = 3x + 4y - 4 = 0$$

Solving Eqs. (i) and (ii)

$$6x + 9y - 3 = 0 \Rightarrow 6x + 8y - 8 = 0 \Rightarrow y = -5 \text{ and } x = 8$$

Point of intersection is $(8, -5)$.

- 23. (d)** $L_1: (1+p)x - py + p(1+p) = 0$

$$L_2: (1+q)x - qy + q(1+q) = 0 \Rightarrow L_3: y = 0$$

Point of intersection: $y = 0$ in L_1

$$x = -p; B(-p, 0) \Rightarrow y = 0 \text{ in } L_2$$

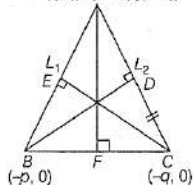
$$x = -q; C(-q, 0)$$



Point of intersection : $y = 0$ in L_1

Point of intersection L_1 and L_2 : $x = pq, y = (1+p)(1+q)$

$A(pq, (p+1)(q+1))$



Altitude from A to BC is $x = pq$

Altitude from B to AC is $y = \frac{-q}{1+q}(x+q)$

Solving the two lines, we get $x = pq$ and $y = -pq$

$\therefore$ Locus of orthocenter is $x + y = 0$.

This is a straight line.

- 24. (d)** Equation of the line perpendicular to $x - 2y = 1$

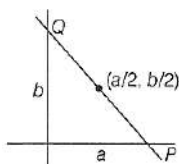
Let equation of line be, $2x + y = \lambda$

It passes through $(1, 1)$: $\lambda = 3$

$\therefore L: 2x + y = 3$

$\Rightarrow y = -2x + 3$

- 25. (b)** Let equation be $\frac{x}{a} + \frac{y}{b} = 1$



$(4, 2)$ lies on $PQ \Rightarrow \frac{4}{a} + \frac{2}{b} = 1$

In the given diagram circumcentre will be $(\frac{a}{2}, \frac{b}{2})$

We have to find locus of circumcentre which we will assume as (h, k)

Hence, $a = 2h, b = 2k$

$\frac{4}{2h} + \frac{2}{2k} = 1 \Rightarrow \frac{2}{h} + \frac{1}{k} = 1$ or $\frac{2}{x} + \frac{1}{y} = 1$

- 26. (b)** $6x^2 - 2xy - 2y^2 = 0$

$m_1 + m_2 = \frac{-2h}{b}, m_1 m_2 = \frac{a}{b}$

$m_1 + m_2 = \frac{2 \times 1}{-2}, m_1 m_2 = \frac{6}{-2}$

$m_1 + m_2 = -1, m_1 m_2 = -3$

$(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1 m_2$

$(m_1 - m_2)^2 = (-1)^2 - 4 \times (-3) = 1 + 12$

$m_1 - m_2 = \sqrt{13} \approx \frac{7}{2}$

- 27. (a)**

$x^2 - 7xy + 12y^2 = 0$

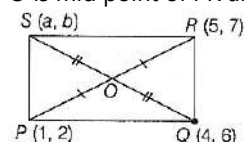
$a = 1, h = -\frac{7}{2}, b = 12$

$\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a+b|} \Rightarrow \tan \theta = \frac{2\sqrt{\frac{49}{4} - 12}}{13} \Rightarrow \tan \theta = \frac{1}{13}$

$\frac{2\cos \theta + 3\sin \theta}{4\sin \theta + 5\cos \theta} = \frac{2+3\tan \theta}{4\tan \theta + 5} = \frac{2+\frac{3}{13}}{\frac{4}{13}+5} = \frac{29}{69}$

- 28. (a)** In a parallelogram diagonals bisect each other.

O is mid point of PR and SQ.



$O(x, y) = (3, \frac{9}{2})$

$\therefore (3, \frac{9}{2}) = (\frac{a+4}{2}, \frac{b+6}{2})$

$a = 2, b = 3$

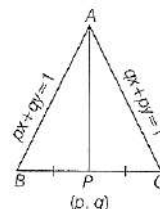
- 29. (d)**

$AB = px + qy = 1$

$AC = qx + py = 1$

$AP: AB + \lambda AC = 0$

i.e. $(px + qy - 1) + \lambda(qx + py - 1) = 0$



Now Eq. (i) passes through (p, q)

$\Rightarrow (p^2 + q^2 - 1) + \lambda(2pq - 1) = 0$

$\lambda = -\frac{(p^2 + q^2 - 1)}{2pq - 1}$

$\therefore AP: (px + qy - 1) - \frac{(p^2 + q^2 - 1)}{2pq - 1}(qx + py - 1) = 0$

$\Rightarrow (2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)(qx + py - 1) = 0$

- 30. (a)** Note that area of the parallelogram bounded by the lines $y = m_1x + c_1, y = m_1x + c_2$ and $y = m_2x + d_1, y = m_2x + d_2$ is given by $\left| \frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2} \right|$

Area of the parallelogram

$y = 4x + 0$

$y = -x + 0$

$c_1 = 0, c_2 = 1$

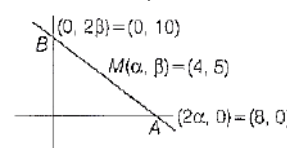
$m_1 = 4,$

$\left| \frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2} \right| = \left| \frac{(0 - 1)(0 - 1)}{4 - (-1)} \right| = \frac{1}{5}$

$\left| \frac{(0 - 1)(0 - 1)}{4 - (-1)} \right| = \frac{1}{5}$

- 31. (d)**

If (α, β) be the mid point of intercepted part AB on axes, then $A(2\alpha, 0), B(0, 2\beta)$ equation of AB will be $\frac{x}{2\alpha} + \frac{y}{2\beta} = 1$. Applying same concept



$\frac{x}{8} + \frac{y}{10} = 1 \Rightarrow 5x + 4y = 40$

- 32. (a)** Distance between parallel lines

$ax + by + c_1 = 0, ax + by + c_2 = 0 \Rightarrow d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

Note For applying this formula coefficient of x and y must be same in both lines.

$-2x + y = 2$ and $2x - y = 2$

$2x - y + 2 = 0$ and $2x - y - 2 = 0$ as we know coefficient of x and y must be same in both lines.

$a = \frac{|2 - (-2)|}{\sqrt{2^2 + (-1)^2}} = \frac{4}{\sqrt{5}}$

$4x - 3y - 5 = 0$ and $8x - 6y + 1 = 0$

$4x - 3y - 5 = 0$ and $4x - 3y + \frac{1}{2} = 0$

$b = \frac{|(-5) - \frac{1}{2}|}{\sqrt{4^2 + (-3)^2}} = \frac{11}{10} \Rightarrow a = \frac{4}{\sqrt{5}}, b = \frac{11}{10}$



$$10b = 11 \Rightarrow 40b = 44$$

$$\Rightarrow 40b = 4 \times 11 \Rightarrow 40b = \sqrt{5}a \times 11$$

- 33. (c)** If it lies in the fourth quadrant, we get $(x, -x)$

Put $y = -x$ in both straight lines.

$$2ax + c = 0 \text{ and } 3bx + d = 0$$

$$\Rightarrow x = -\frac{c}{2a} \text{ and } x = -\frac{d}{3b} \Rightarrow \frac{c}{2a} = \frac{d}{3b} \Rightarrow 3bc - 2ad = 0$$

- 34. (b)** $L: x + y - 1 = 0$

As $(3, 2)$ and $(\cos \theta, \sin \theta)$ lies on same side of line

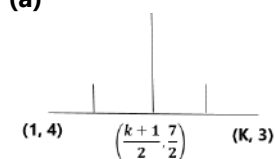
$$L: (3, 2)3 + 2 - 1 = 4 > 0$$

$$\text{So, } L: (\cos \theta, \sin \theta) \equiv \cos \theta + \sin \theta - 1 > 0$$

$$\Rightarrow \sqrt{2} \left[\frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta \right] > 1 \Rightarrow \sin \left(\theta + \frac{\pi}{4} \right) > \frac{1}{\sqrt{2}}$$

$$\Rightarrow \frac{\pi}{4} < \theta + \frac{\pi}{4} < \frac{3\pi}{4} \Rightarrow 0 < \theta < \frac{\pi}{2}$$

- 35. (a)**



Eq. of perpendicular bisector

$$\left(y - \frac{7}{2} \right) = \left(\frac{k-1}{3-4} \right) \left(x - \frac{K+1}{2} \right)$$

$$y - \frac{7}{2} = (K-1) \left(x - \frac{K+1}{2} \right)$$

for y intercept $x = 0$

$$y - \frac{7}{2} = (K-1) \left(x - \frac{K+1}{2} \right)$$

$$\frac{2y-7}{2} = \frac{(1-K)(1+K)}{2}$$

$$2y-7 = 1-K^2$$

$$K^2 = 8 - 2 \times |4|$$

$$= 8 - 8 = 0 \text{ or } 8 + 8 = 16$$

$$K = 0 \text{ and } K = \pm 4$$

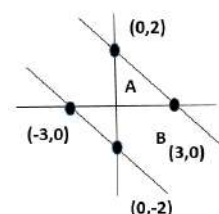
- 36. (a)** The point $(1, 1/2)$ and $(3, -1/2)$ are

$$\text{Lines} = 2x + 3y = 6$$

$$= 2x + 3y = -6$$

$$A = (1, 1/2)$$

$$B = (3, -1/2)$$



Between the lines.

- 37. (a)** $6x^2 - xy + \lambda y^2 = 0$

$$a = 6, b = \lambda, h = -1/2$$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

for perpendicular line $a+b = 0$

$$6 + \lambda = 0$$

$$\lambda = -6$$

Of inclined at an angle of π

$$\tan \frac{\mu}{4} = \frac{2\sqrt{\frac{1}{4} - 6\lambda}}{6+\lambda}$$

$$6 + \lambda = \frac{2\sqrt{1-24\lambda}}{2}$$

Squaring both side

$$(6+\lambda)^2 = (1-24\lambda)$$

$$\lambda^2 + 36 + 12\lambda = 1 - 24\lambda$$

$$\lambda^2 + 35 + 36\lambda = 0$$

$$(\lambda + 35)(\lambda + 1) = 0$$

$$\lambda = -35 \text{ or } \lambda = -1$$

- 38. (a)**

Lines:

$$2y = x + 1 \rightarrow m_1 = \frac{1}{2}$$

$$y = 3x + 2 \rightarrow m_2 = 3$$

Use formula:

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{3 - \frac{1}{2}}{1 + \frac{3}{2}} \right| = \left| \frac{5/2}{5/2} \right| = 1$$

$$\theta = \tan^{-1}(1) = \frac{\pi}{4} \text{ (acute angle)}$$

$$\text{Obtuse angle} = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\text{Ans. } \frac{3\pi}{4}$$

- 39. (b)**

Find intercepts quickly:

$$\frac{x}{4} + \frac{y}{2} = 1$$

$$A = (4, 0), B = (0, 2)$$

midpoint:

$$M = \left(\frac{4+0}{2}, \frac{0+2}{2} \right) = (2, 1)$$

Reflection across $x + y = 1$:

Shortcut for reflection over $x + y = 1$; swap slope along perpendicular and shift:

$$M' = (0, -1)$$

P on $x + y = 1$ with $AP \perp BP$:

$$\text{Let } P = (t, 1-t), \text{ solve } AP^2 - BP^2 \rightarrow \text{QUICK: } 2T^2 - 10T + 17 + 2T + 1$$

$$t = \frac{4}{3}$$

$$\text{So } P = \left(\frac{4}{3}, -\frac{1}{3} \right)$$

distance $M'P$:

$$d = \sqrt{(4/3 - 0)^2 + (-1/3 + 1)^2}$$

$$= \sqrt{16/9 + 4/9}$$

$$= \frac{2\sqrt{5}}{3}$$